

DSA 8401: Applied Machine Learning

Assignment 2 — The Cost of Being Wrong

September 12, 2026

Abstract

This study evaluates cost-sensitive fraud classification on the unchanged output of Assignment 1. Its dataset uses `is_fraud` as its target which was retained to preserve the required A1 modeling output. The cleaned dataset contains 180,000 transactions, 3,991 customers and an 8.41% fraud rate. A strict temporal evaluation design was used, comprising an untouched recent holdout, a seven-day purge interval, and five expanding-window purged cross-validation folds. Regularized logistic regression, random forest and LightGBM were compared using PR-AUC as the principal metric. Optuna tuning improved LightGBM to a development PR-AUC of 0.141 ± 0.003 . Platt calibration was subsequently applied and the operating threshold was selected using asymmetric costs of KES 10,000 for a false negative and KES 800 for a false positive. On the untouched out-of-time holdout, the final model achieved PR-AUC 0.143, ROC-AUC 0.641 and a Brier score of 0.0756. The selected threshold of 0.0784 reduced estimated cost by approximately KES 234,548 per 1,000 transactions relative to the conventional 0.50 threshold.

1 Evaluation Design

1.1 Data and target

The Assignment 1 modelling table was used without altering its observations or engineered feature set, as required by the assignment. It contained 180,000 transactions from February 2025 to July 2026, covering 3,991 customers. The positive class was `is_fraud=1`, with an empirical prevalence of 8.41%. Known post-outcome leakage fields, including `manual_review_score` and `settlement_status`, remained excluded from the modelling features.

The assignment specifies unequal error costs. These were mapped to the available fraud target: a missed fraudulent transaction (false negative) costs KES 10,000, while incorrectly flagging a legitimate transaction (false positive) costs KES 800. The decision cost is therefore

$$C(t) = 10,000 FN(t) + 800 FP(t).$$

1.2 Temporal testing protocol

Random cross-validation would allow information from later periods to influence estimates for earlier periods. The data were therefore sorted chronologically and the most recent approximately 15% of observations were reserved as an untouched final holdout. The development period ended on 4 May 2026, a seven-day purge interval followed, and the holdout covered 11 May–31 July 2026. This produced 150,700 development observations and 27,000 holdout observations.

All development experiments used the same five expanding-window purged time-series folds, with a seven-day gap between each training period and its validation block. Preprocessing was fitted separately inside every training fold. The holdout was not used for model comparison, hyperparameter tuning, calibration selection or threshold optimisation.

PR-AUC was selected as the headline ranking metric. With only about 8.4% positives, the PR-AUC baseline is approximately 0.084. ROC-AUC was retained as a secondary diagnostic, but can appear reassuring under imbalance because the large number of negatives dominates the false-positive-rate denominator. PR-AUC more directly measures the precision–recall trade-off relevant to finding rare fraud cases.

2 Model Comparison and Hyperparameter Tuning

Three model families were evaluated on identical temporal folds: L2-regularised logistic regression, random forest and LightGBM. Logistic regression provides a high-bias, low-variance linear baseline; random forest uses bagging and feature subsampling primarily to reduce tree variance; boosting sequentially corrects residual errors and can reduce bias, although it is more sensitive to regularisation and hyperparameters.

Table 1: Purged temporal cross-validation performance on development data.

Model	PR-AUC	SD	ROC-AUC	Brier
Tuned LightGBM	0.1406	0.0034	0.6327	0.0750
Random Forest	0.1340	0.0039	0.6249	0.0752
Logistic Regression	0.1332	0.0048	0.6238	0.0753
LightGBM, untuned	0.1317	0.0066	0.6179	0.0756

The untuned random forest narrowly led the initial comparison, but the differences among the three baselines were small. Untuned LightGBM also showed the greatest fold-to-fold variation. Hyperparameter optimisation was therefore applied to LightGBM using Optuna with a fixed budget of 60 trials and mean purged-CV PR-AUC as the objective.

The search covered learning rate [0.01, 0.20] on a log scale, 15–127 leaves, maximum depth 3–10, minimum child samples 20–300, row and feature subsampling 0.60–1.00, and log-scaled L1/L2 regularisation. A TPE sampler and **MedianPruner** were used; 38 trials completed and 22 were pruned. The best trial obtained mean PR-AUC 0.1406 with learning rate 0.0144, depth 3 and minimum child support 234. The shallow, low-learning-rate configuration suggests that stronger regularisation was beneficial. Tuning improved both mean ranking performance and temporal stability, making tuned LightGBM the development champion.

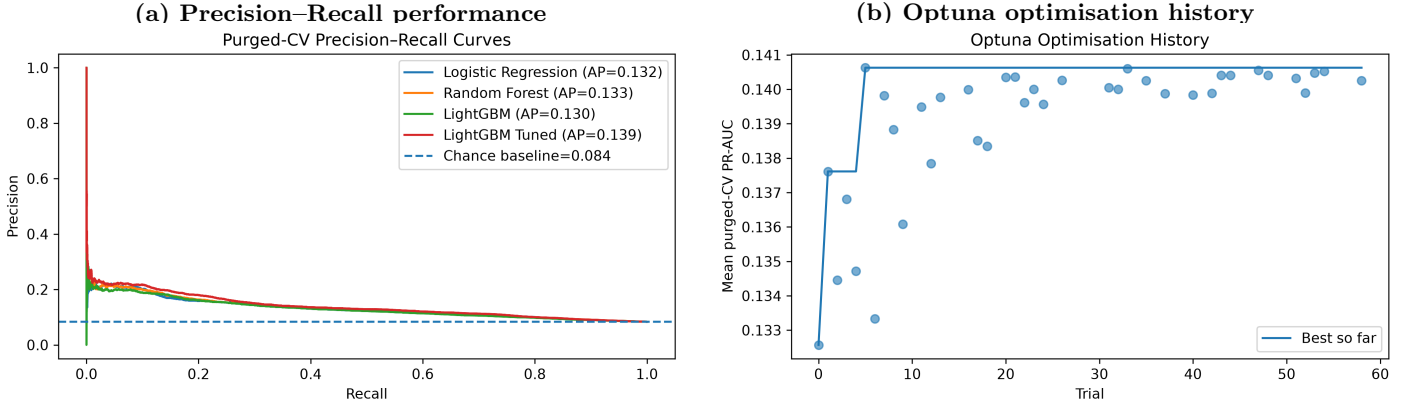


Figure 1: Development-set ranking performance and budget-limited hyperparameter optimisation.

3 Imbalance, Calibration and Cost-Sensitive Decisions

3.1 Imbalance treatment

Two imbalance remedies were compared with plain logistic regression. Class weighting preserved almost the same ranking performance (PR-AUC 0.1327 versus 0.1332) but increased the Brier score from 0.0753 to 0.2387, indicating severely distorted raw probability estimates. SMOTE-NC performed worse, reducing mean PR-AUC to 0.1179 and ROC-AUC to 0.6041. Consequently, neither method was retained; the original class distribution was preserved and the asymmetric business problem was handled primarily at the decision-threshold level.

A deliberate anti-pattern experiment also demonstrated why resampling must occur inside training folds. Correct inside-fold oversampling produced PR-AUC 0.1345 ± 0.0033 , while resampling before cross-validation produced an apparently much higher 0.3328 ± 0.0034 . However, the invalid procedure changed the evaluated positive prevalence from 8.4% to 23.1%, while ROC-AUC remained almost unchanged (0.6242 versus 0.6257). Thus the apparent PR-AUC gain was an evaluation artefact rather than evidence of substantially improved discrimination.

3.2 Probability calibration

Calibration was performed using honest out-of-fold predictions from development data. Platt scaling and isotonic regression were compared on a later temporal calibration-selection window; Platt scaling achieved the lower Brier score and was retained. On the subsequent development decision window, the Brier score changed only from 0.07437 to 0.07436, indicating that the tuned model was already reasonably calibrated and required only a small probability-scale correction. Calibration nevertheless moved the empirical cost-minimising threshold from 0.0800 to 0.0784. For reference, the theoretical threshold for perfectly calibrated probabilities under the specified costs is

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{800}{800 + 10,000} = 0.0741.$$

The empirical calibrated threshold of 0.0784 was selected on development data and frozen before opening the final holdout.

3.3 Final cost evaluation

The untouched holdout retained a fraud prevalence of 8.44%. Raw tuned-LightGBM performance was PR-AUC 0.1431, ROC-AUC 0.6408 and Brier 0.07559; after Platt calibration the ranking metrics were unchanged and Brier improved marginally to 0.07558.

At threshold 0.0784 the holdout confusion matrix contained 13,461 true negatives, 11,259 false positives, 746 false negatives and 1,534 true positives. In contrast, the 0.50 threshold identified no positive cases in the holdout, causing all 2,280 actual fraud cases to become costly false negatives. This illustrates why ranking performance and the operational cut-off must be treated as separate decisions.

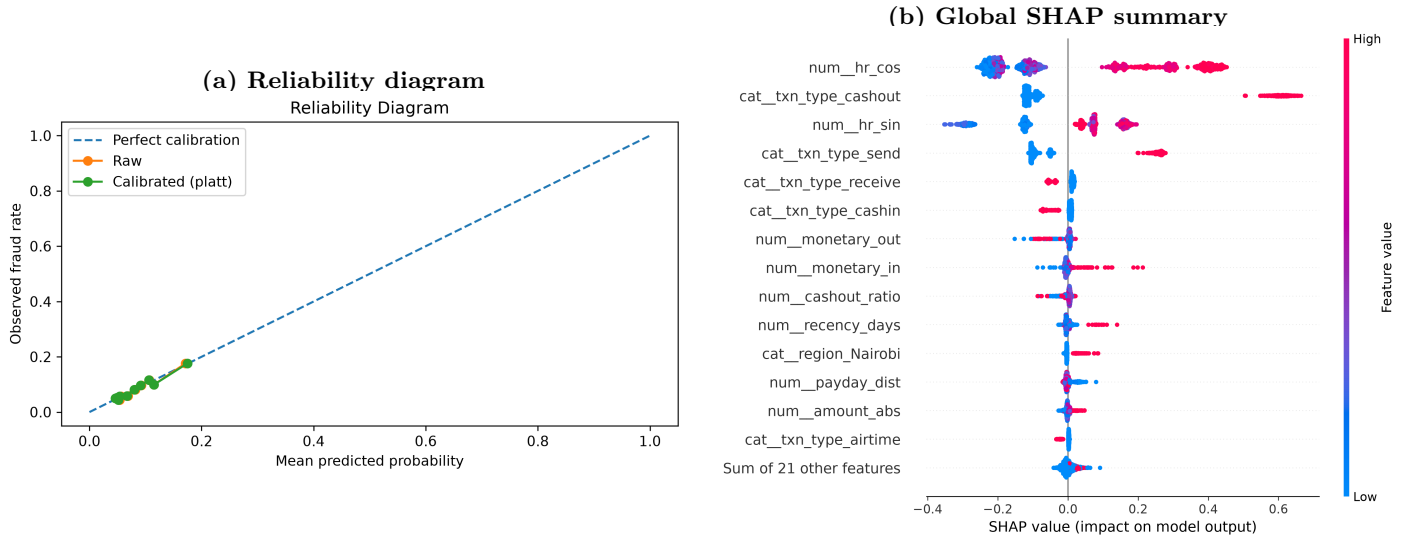


Figure 2: Probability calibration and global model explanation.

Table 2: Final holdout cost comparison per 1,000 transactions.

Decision rule	Threshold	Cost / 1,000	Saving vs. 0.50
Selected development rule	0.0784	KES 609,896	KES 234,548
Theoretical calibrated rule	0.0741	KES 612,593	KES 231,852
Naive rule	0.5000	KES 844,444	—

4 Explainability, Fairness and Regulatory Considerations

Global and local SHAP explanations were generated for the final LightGBM model. The global beeswarm plot identifies the variables that most consistently move predicted fraud risk, while the local waterfall explanation decomposes one individual prediction into feature-level positive and negative contributions. The local explanation is retained with the reproducible notebook and model artifacts.

Fairness was evaluated across two operationally meaningful subgroup dimensions: **region** and **segment**. The assignment’s “approval rate” was operationalised as the *not-flagged rate*, i.e., the proportion of transactions classified as legitimate at the selected threshold. The false-negative rate (FNR) measures the proportion of actual fraud cases that the model failed to flag.

Table 3: Subgroup fairness metrics at the selected threshold of 0.0784.

Region				Segment			
Group	<i>n</i>	Not-flagged	FNR	Group	<i>n</i>	Not-flagged	FNR
Mombasa	2,762	48.3%	30.5%	Retail	20,173	51.7%	32.4%
Jinja	2,713	54.5%	36.2%	Merchant	3,661	53.0%	35.9%
Kigali	2,680	51.6%	35.0%	Agent	1,283	57.4%	33.9%
Nairobi	2,610	50.3%	33.5%				
Mwanza	2,467	51.8%	38.5%				
Arusha	2,427	55.3%	35.0%				
Dar es Salaam	2,402	52.2%	29.2%				
Eldoret	2,395	51.6%	28.0%				
Kampala	2,345	52.2%	30.8%				
Nakuru	2,316	54.7%	33.7%				

The subgroup results reveal greater variation by region than by customer segment. Regional not-flagged rates range from 48.3% in Mombasa to 55.3% in Arusha, a 7.0 percentage-point spread. More importantly, regional FNR ranges from 28.0% in Eldoret to 38.5% in Mwanza, a 10.5 percentage-point difference. This means that, at the same global threshold, the model misses materially different proportions of actual fraud across regions. Segment-level differences are smaller: not-flagged rates range from 51.7% to 57.4%, while FNR ranges from 32.4% to 35.9%. These results do not by themselves establish unlawful bias, because subgroup base rates and feature distributions may differ, but they identify regional disparities that require investigation and continuing monitoring before deployment.

The Central Bank of Kenya Digital Credit Providers Regulations, 2022 establish consumer-protection, complaints-resolution and responsible-use requirements for customer and credit information [1]. They do not prescribe a specific SHAP-style explanation format for every automated rejection. The practical regulatory expectation is therefore *traceability and defensibility*: the provider should be able to identify the decision rule applied, the relevant customer information used, and the material factors that led to the rejection. For a production credit decision system, SHAP-derived reason codes should therefore be translated into clear customer-facing explanations, with records retained for complaints, review

and model governance. Fairness metrics should also be monitored after deployment rather than treated as a one-off pre-release test.

5 Reproducibility

The complete workflow is packaged in the `a2-models` repository. Saved artifacts include fold-level and model-comparison tables, development out-of-fold predictions, final holdout predictions, calibration and cost tables, fairness tables, PR and reliability curves, SHAP figures, the Optuna SQLite study database, the fitted LightGBM pipeline, and the serialized calibration and threshold object. Fixed random seeds and relative project paths allow a fresh clone, supplied with the unchanged A1 modeling artifact, to reproduce the comparison table and final decision outputs.

Deployment Recommendation

Tuned LightGBM with Platt calibration and a single fraud-flag threshold of 0.0784 is recommended for deployment. It achieved the strongest temporal generalisation, with holdout PR-AUC 0.1431 and ROC-AUC 0.6408, while preserving acceptable probability calibration. Most importantly, the fixed 0.0784 threshold reduced estimated holdout loss by approximately **KES 234,548 per 1,000 transactions** relative to the conventional 0.50 rule. The threshold should be deployed together with ongoing calibration, subgroup false-negative-rate and cost monitoring rather than treated as permanently fixed.

References

- [1] Central Bank of Kenya. *The Central Bank of Kenya (Digital Credit Providers) Regulations, 2022*, Legal Notice No. 46 of 2022.
- [2] Serrano, J. and Bundi, E. L. *Applied Machine Learning Course Notes: Classical Machine Learning, Rigorous Evaluation, Ensembles and Hyperparameter Tuning*. Strathmore University, 2026.